Soft Computing (2024) 28:8715–8730
https://doi.org/10.1007/s00500-023-08973-5

APPLICATION OF SOFT COMPUTING

Portfolio dynamic trading strategies using deep reinforcement learning

Min-Yuh Day[1] · Ching-Ying Yang[1] · Yensen Ni[2]

Accepted: 1 July 2023 / Published online: 30 July 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract
Using the constituent stocks of the iShares MSCI US ESG Select Index ETF, a matrix of technical indicators, returns, and covariance is incorporated to represent the inherent information characteristics of the stock market. In this study, based on the proposed Deep Reinforcement Learning for Portfolio Management on Environmental, Social, and Governance (DRLPMESG) architecture model, investors who use active portfolio management reap the greatest rewards, as the portfolio with 5 stocks performing the best, with an annualized return of 46.58%, a Sharpe ratio of 1.37, and a cumulative return of 115.18%, indicating that the results have the potential to win the market and generate excess profits. In contrast to the efficient market hypothesis, this new understanding of proven effectiveness in obtaining satisfactory rewards would help improve investment strategies for portfolio management. Furthermore, this study proposed that holding 5 stocks in a portfolio can lead to higher returns, laying the foundation for future research on the number of holdings. Moreover, when compared to previous static strategies, this model offering a dynamic strategy may generate a more stable return in the face of market fluctuations.

Keywords Artificial intelligence · Deep reinforcement learning · Financial technology · Portfolio management · Trading strategy

1 Introduction

Modern portfolio management theory seeks long-term, stable profits with manageable risks through asset allocation, portfolio, and investment strategy. Passive and active portfolio management techniques are used in investment management (Al-Aradi and Jaimungal 2021; Da Silva et al. 2009; Ieda et al. 2019; Peterson 2012; Stoilov et al. 2021). The formation of market portfolios for the use of index funds purchased in various asset markets or portfolios based on the efficiency frontier are examples of passive investment management (Markowitz 1952). However, active portfolio management primarily employs one of two approaches (entrusting operations to professional managers or designing a strategy on one's own) to improve market performance and obtain excess returns.

Artificial intelligence techniques have been widely used in the financial sector to perform different tasks (Ahmed et al. 2022). Among them, deep learning has been received increasing attention in finance (Ozbayoglu et al. 2020). RNN, DMLP, and CNN models[1] are often implemented in deep learning. Algorithmic trading, financial sentiment analysis and behavioral finance, and financial text mining are commonly used in the field of finance. Regarding reinforcement learning, a kind of machine learning, agents will constantly interact with their environment and learn to perform tasks correctly through trial and error. The time-delayed learning mechanism combined with cumulative

✉ Yensen Ni
ysni@mail.tku.edu.tw; ysniysni@gmail.com

Min-Yuh Day
myday@gm.ntpu.edu.tw

[1] Graduate Institute of Information Management, National Taipei University, Taipei, Taiwan

[2] Department of Management Sciences, Tamkang University, No.151, Yingzhuan Rd., Tamsui Dist., New Taipei City 25137, Taiwan

[1] In deep learning, Recurrent Neural Network (RNN), Deep Multi-layer Perceptron (DMLP), and Convolutional Neural Network (CNN) models are frequently used.

rewards in reinforcement learning makes it natural to consider decision making (Sutton and Barto 1998).

According to previous studies, reinforcement learning is more beneficial in the field of portfolio management (Lim et al. 2022; Shi et al. 2022; Wu et al. 2021, 2020; Yu et al. 2023), as portfolio management can be viewed as an optimized process from the outset of asset allocation selection, portfolio decision-making, risk assessment, and providing the best asset action plan (Théate and Ernst 2021). The investment results would be also influenced by time period, risk, market liquidity, correlation, expected return, number of holding shares, and other factors, of which reinforcement learning has adaptive and automated trading characteristics, making it easier to adapt to a complex and volatile market environment.

Based on existing research, the scope of deep learning is extended to reinforcement learning, employing the properties of neural networks to provide the best features through deep and complicated learning, so that agents can better replicate and simulate human thinking and perform (Charpentier et al. 2023; Wang et al. 2020). However, according to the efficient market hypothesis, investors may not make excess profits by outperforming the market because they might not earn excess profits through investment. As a result, investors may profit from passive investment strategies, which are expected to outperform active strategies.

Reinforcement learning is frequently used in financial signal representation (Deng et al. 2017; Taghian et al. 2022), trading (Chen and Huang 2021; Zhang et al. 2020), and portfolio management (Betancourt and Chen 2021; Ozbayoglu et al. 2020). Portfolio research, for example, focuses primarily on risk diversification, implying that holding more shares can effectively reduce risk. However, from the perspective of market transaction, the number of shares held and other factors would have a significant impact on overall profitability (Meng and Khushi 2019), which has received little attention in the relevant literature. As such, this study not only combines portfolio theory with the application of deep reinforcement learning models to obtain the best investment portfolio but also investigates how many stocks can be included to form the best investment portfolio.

The purpose of this study is to explore how deep reinforcement learning models can be used to dynamically optimize portfolios and ensure that investors can not only achieve stable returns but also outperform the market even in volatile stock markets. Furthermore, we consider how many stocks should be included in the optimum portfolio.

In this study, we illustrate several essential findings. First, according to the benchmark method, seven out of nine portfolio groups beat the market at a winning rate of 78% in this study's experimental results. This finding confirmed that most agents can outperform the market and provide higher returns. Second, we also find that self-designed active portfolio management strategies outperform passive portfolio management strategies. For example, SAC agent outperforms 20-stock portfolios with a cumulative rate of return of 115.18%, while A2C agent outperforms 5-stock portfolios with an annualized rate of return of 46.58% and a Sharpe ratio of 1.37, respectively. Third, while investigating how many stocks to hold in a portfolio to attain the best performance, we discovered that a portfolio consisting of five stocks has the best performance.

In addition, we argue that this study can contribute to the existing literature in a variety of ways. First, we show that portfolio management based on deep reinforcement learning can outperform the market, which may contradict the efficient market hypothesis. Second, when compared to previous static strategies, this study offering a several dynamic portfolio management strategies may not only generate a more stable return in the face of market fluctuations, but also proven effectiveness in obtaining satisfactory rewards, which would aid in improving portfolio management. Third, this study suggested that having 5 stocks in a portfolio can result in higher returns, laying the foundation for future research on the number of stocks held in a portfolio. As such, we argue that all of the above may contribute to the existing portfolio management literature.

The remainder of this paper is organized as follows. The literature review is presented in Sect. 2. Section 3 describes the data and methodology used in this study. Section 4 shows the experimental results. Section 5 provides the concluding remarks.

2 Literature review

Portfolio management, including portfolio optimization, selection, and allocation, is the process of selecting various assets in a portfolio. Portfolio management is an optimization problem in which the best options are determined and the best asset action plan is provided. In addition, based on the competition between mutual funds, deep learning has been used for financial decision-making and improving portfolio performance in recent years (Chen et al. 2021; Chen and Ren 2021; Craja et al. 2020; DeMiguel et al. 2021; Vo et al. 2019; Xu et al. 2020).

2.1 Portfolio management

There are two main types of investment management systems for portfolio management: passive and active portfolio management. In investing and holding shares, passive portfolio management strategies primarily replicate a certain market index or benchmark in order to attain a similar

return outcome to the market index or benchmark. Market portfolios formed by purchasing index funds in various asset markets or investment portfolios formed on the basis of efficiency front are used based on the static strategy employed by passive portfolio management (Markowitz 1952). Active portfolio management strategies can be divided into two categories: entrusting operations to professional managers and employing a self-design strategy to obtain market rewards. Such strategies seek to outperform the market based on a certain benchmark (e.g., the market performance).

Passive investment strategies generate returns that are comparable to a particular market index or benchmark. For example, each stock in the index is weighted, and a portfolio weighted by market capitalization represents the market. Active portfolio management is an investment strategy that implements investment decisions based on returns, risks, benchmarks, preferences, and information ratios. All of these variables would influence the expected return in actively managing portfolio (Al-Aradi and Jaimungal 2021; Ieda et al. 2019; Peterson 2012). As such, because the goal of active investment strategies is to outperform the market, they carry more market risk than passive portfolio management strategies (Alexander 2008).

2.2 Deep learning applied to portfolio management

Previous research employed a deep learning model with time series characteristics to forecast the next day or week's results, and then classified the outcomes of short and long trading strategies to optimize the portfolio (Aggarwal and Aggarwal 2017; Ma et al. 2021; Pham et al. 2021; Torrente and Uberti 2022; Vo et al. 2019; Weng et al. 2020; Xu et al. 2020; Yun et al. 2020). In addition, Liang et al. (2018) proposed a deep reinforcement learning model to maximize reward based on the optimal strategy, with the annual reward and Sharpe ratio as the evaluation method. Furthermore, deep learning-based cryptocurrency portfolio management has received a lot of research attention (Liu et al. 2021; Lucarelli and Borrotti 2020). Lucarelli et al. (2020) proposed a deep Q-learning portfolio management framework for a portfolio of four cryptocurrencies. Liu et al. (2021) propose a framework based on a deep reinforcement learning algorithm proximal policy optimization (PPO) for automatic high-frequency cryptocurrency transaction strategy construction.

2.3 Deep reinforcement learning

Machine learning mainly includes supervised, semi-supervised, unsupervised, and reinforcement learning (Ahmed et al. 2022). Deep learning, a branch of machine learning, could be added to the field of machine learning in the task of artificial intelligence application (Castiglioni et al. 2021). Reinforcement learning is a type of machine learning, in which agents constantly interact with their surroundings and learn to perform a task correctly through trial and error. Deep reinforcement learning is a combination of deep learning and reinforcement learning that agents can use neural network characteristics in deep learning to learn the maximum reward function or the optimal strategy (Arulkumaran et al. 2017).

2.4 Deep reinforcement learning in portfolio management

Because deep reinforcement learning has recently been employed in portfolio management, we have included several relevant papers in Table 1. According to the review, the majority of research directions on deep reinforcement learning in portfolio management employ value-based models.

Table 1 shows that calculus methods based on DQN, Q-learning, and other value functions are widely proposed in relevant studies (Darapaneni et al. 2020; Huang et al. 2020; Rundo 2019; Wu et al. 2020). Furthermore, while Zhang et al. (2020), Liu et al. (2021), and Huang et al. (2020) use the dominant A2C, PPO, and DDPG models, recent studies have increasingly used actor-critic models (A2C) as calculus methods, as well as common evaluation methods, such as annualized return, Sharp ratio, and maximum drawdown.

3 Data and methodology

Deep reinforcement learning models, the majority of which are based on critic-only models, have been used to train financial agents. However, in recent years, the critic-actor model has outperformed the critic-only model. Furthermore, when investing in the stock market, investors may use sustainability indicators (e.g., technical indicators). As a result, different deep reinforcement learning models with actors and critics would be frequently used to optimize portfolio investing strategies while also taking sustainability indicators into account for asset allocation.

3.1 Research design

On the basis of portfolio management theory, investment management strategies are divided into passive and active portfolio management. In the active portfolio management

Table 1 Review of the literature on deep reinforcement learning in portfolio management

References	Data set	Data period	Features setting	Deep learning method	Performance criteria	Transaction cost
Huang et al. (2020)	CSI300 index, 3 stocks in CSI500	2007–2020	OCHLV	DDPG	Average Return: 0.000949 Annual Sharpe: 1.707 Annual Sortino: 2.946 MDD: 8.09%	Yes, 0.0025
Zhang et al. (2020)	50 futures contracts	2005–2019	OCHLV, MA, MACD, RSI	LSTM, DQN, PG, A2C	$E(R)$: 1.258 Std.(R): 0.976 Sharpe: 1.288, Sortino: 2.220 MDD: 0.002	Yes, 0.0001
Rundo (2019)	EUR/USD	2004–2018	OCHLV, technical indicators	LSTM, DQN	ROI Max = 98.81% MD Max = 19.28%	No
Liu et al. (2021)	Bitcoin	2017–2020	OCHLV, technical indicators	LSTM, PPO	Profits rate = 1.46%	Yes, 0.0025
Darapaneni et al. (2020)	16 index from Indian market	2011–2019	OCHLV, SMA	Q-learning	Annualized return = 8.54% Max drawdown = 1.44%	Yes, 0.0030
Wu et al. (2020)	3 stocks from US, 3 stocks from UK, and 3 stocks from China	2008–2018	OCHLV, MA, EMA, ACD, BIAS	GRU, DQN, DPG	Return = 215.9% SR = 1.9%	No

In deep learning, recurrent neural network (RNN), deep multilayer perceptron (DMLP), and convolutional neural network (CNN) models are frequently used. Besides, deep deterministic policy gradient (DDPG), deep Q-network (DQN), Advantage Actor-Critic (A2C), and deterministic policy gradient (DPG) are all deep leaning methods. OCHLV datasets consist of the opening price, highest price, lowest price, closing price, and trade volume of a specific timeframe

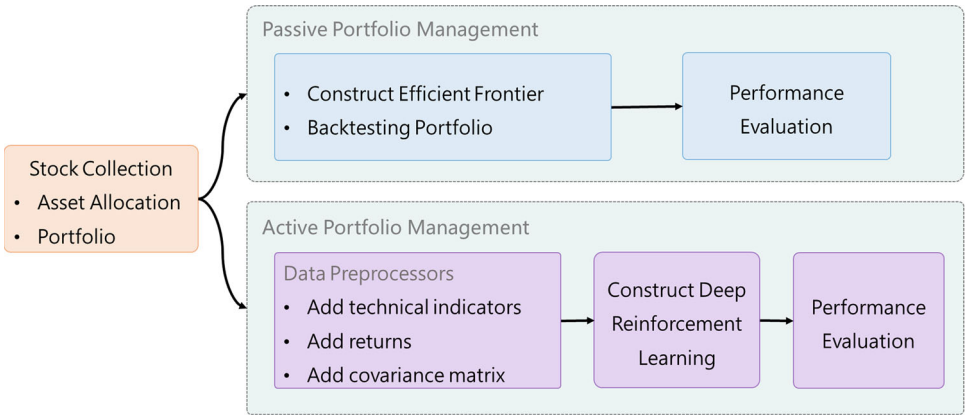


Fig. 1 Research framework of deep reinforcement learning in portfolio management

strategy, the deep reinforcement learning model is used to optimize the investment trading portfolio strategy. Figure 1 depicts the deep reinforcement learning research framework in portfolio management, which includes performance evaluation.

3.2 Data collection procedure

In the post-epidemic era, factors such as sustainability, financial stability, and a strong management team have become important investment directions for investors since

investors are increasingly using the ESG evaluation indicator to evaluate companies (Kocmanová and Dočekalová 2012). The ESG evaluation indicator is a corporate social responsibility concept that links investment performance to environmental, social, and governance indicators. ESG is employed as a guide and investment strategy for business management in order to achieve the financial goal of chasing long-term profits while being willing to take risks (Escrig-Olmedo et al. 2010), helping companies to select portfolios that include sustainability indices. As a result, the data for this study comes from the iShares MSCI American ESG Selected Index ETF, an influential ESG sustainability index that has been around for a long time.

3.2.1 Datasets

The iShares MSCI US ESG Select Index, which has 227 constituent stocks (companies), was used in the study. Each company performs well in corporate responsibility, the social environment, and social welfare. These constitute stocks could be classified into 12 categories (industries), including information technology stocks, health care stocks, financial stocks, etc.

In addition, we adopt the daily data from Jan. 2000 to Dec. 31, 2020, a total of 20 years of daily trading data for these constituent stocks (i.e., 1,003,698 samples in total) as our investigated samples. We then collect trading volume, open price, high price, low price, close prices, and adjust closing prices for each sample.

Furthermore, because the ETF composition is calculated as a capitalization-weighted index, in which index components with higher market capitalization receive more weight in the index, companies with lower market capitalization have less influence on the overall index, which is used to make the ETF close to the market index. To ensure that ETFs are similar to market indices, the market capitalization weighted index is used as a calculation method for ETFs.

In this study, we use iShares MSCI US ESG Select Index ETF data. The framework treats portfolio weights as actions and returns as rewards, representing the intrinsic information characteristics of the stock market by incorporating a matrix of technical indicators, returns, and covariances for building dynamic portfolios based on various deep reinforcement learning models.

3.2.2 Data cleaning, feature selection, and data splitting processes

In terms of data management, we first clean the data by replacing the original closing price with the adjusted closing price and deleting null values from the dataset in the data clean process. As model inputs, we include six technical indices (moving average, relative strength index, commodity channel index, average directional indicator, Bollinger bands, moving average convergence and divergence), two returns (daily return, log return), and a covariance matrix in addition to stock price in the feature selection process. Regarding data splitting process, the data set with 20 years of data is divided into 90% and 10% for the purpose of obtaining additional data for training. As a result, 18 years of data serve as the training set, and 2 years serve as the test set.

3.3 Asset allocation and portfolio evaluation

The portfolio's premise is that specific assets that will demonstrate satisfactory performance are unknown, and whether history will repeat itself is unknown. As a result, we will avoid selecting all markets that have historically performed well and will instead design the portfolio using the asset allocation principle. Risk, period, correlation, expected return, and holding quantity are all factors considered in asset allocation and portfolio evaluation to minimize risk and maximize return. The portfolio's primary goal is to diversify industries, allocate assets to various industries, and reduce risks (Choueifaty and Coignard 2008).

Evans and Archer (1968) pointed out that the risk of a portfolio built by picking 15 stocks at random will not be higher than the overall market risk. As per investment research and portfolio management theory (Chandra 2017), a single portfolio with 12 to 18 stocks can provide about 90% of investment diversification benefits. According to the number of shares held in earlier studies, the number of shares held is divided into 10 and 20 stocks as portfolios. In a prior study, a different number was utilized, and a portfolio with only five files was included to discuss how many portfolios demonstrate optimal results.

3.3.1 Asset allocation

In this study, constituent stocks of the iShares MSCI US ESG Select Index ETF are chosen at random to form the optimal portfolio, and stocks from 12 industries are selected randomly to ensure that each category contains a maximum of two stocks. The combination with low correlation is determined by the risk management assessment method. The relationship between expected stock return and risk is examined, and a portfolio of risk-free and risk-adjusted assets is constructed using the capital asset pricing model (CAPM) (Lintner 1965). To determine which deep reinforcement learning-based model performs best, a different number of portfolios are used.

This study's experimental models include portfolios of 5, 10, and 20 stocks from various industries. There are nine

Table 2 Advantage Actor-Critic (A2C) algorithm pseudocode

Algorithm 1 Advantage Actor-Critic (A2C)

//Assume global shared $\theta, \theta^{-}, and\ counter\ T = 0$.
Initialize thread step counter $t \leftarrow 0, \theta^{-} \leftarrow \theta, d\theta \leftarrow 0$.
Get initial state s.
repeat
  Take action a with ϵ-greedy policy base on $Q(s, a;\ \theta)$
  Receive new state s' and reward r
  $$y = \begin{cases} r & for\ terminal\ s' \\ r + \gamma max_{a'} Q(s', a'; \theta^{-}) & for\ non-terminal\ s' \end{cases}$$
  $s = s'$
  $T \leftarrow T + 1\ and\ t \leftarrow t + 1$
  If $T\ mod\ I_{target} == 0$ **then**
    Update the target network $\theta^{-} \leftarrow \theta$
  end if
  if t $mod\ I_{AsyncUpdate} == 0$ or s is terminal **then**
    Perform asynchronous update of θ using $d\theta$.
    Clear gradients $d\theta \leftarrow 0$.
  end if
until $T > T_{max}$

Table 3 Proximal policy optimization (PPO) algorithm pseudocode

Algorithm 2 Proximal Policy Optimization (PPO)

Initialize actor $\mu: S \rightarrow R^{m+1}$ and $\sigma: S \rightarrow diag(\sigma_1, \sigma_2, \ldots, \sigma_{m+1})$
for i = 1 to M **do**
  Run policy πθ ~ N(μ(s), σ(s)) for T timesteps and collect (s_t, a_t, r_t)
  Estimate advantages $\hat{A}_t = \sum_{t'>t} \gamma^{t'-t} r_{t'} - v(s_t)$
  Update old policy $\pi_{old} \leftarrow \pi_0$
  for j = 1 to N **do**
    Update actor policy by policy gradient:
    $$\sum_i \nabla_\theta L_i^{CLIP}(\theta)$$
    Update critic by:
    $$\nabla L(\phi) = -\sum_{t=1}^{T} \nabla \hat{A}_t{}^2$$
  end for
end for

portfolios in the trained model. Portfolios with 20 stocks are labeled 20-1, 20-2, and 20-3. Portfolios with ten stocks are labeled 10-1, 10-2, and 10-3. Portfolios with five stocks are labeled 5-1, 5-2, and 5-3.

3.3.2 Portfolio evaluation

Portfolio management aims to identify and implement a set of investments that meet long-term financial objectives and risk tolerance. As a result, understanding the correlation between various assets is necessary in order to effectively reduce risks and achieve the risk management goal. According to the capital asset pricing model, a two-dimensional heatmap matrix is used to show whether each portfolio exhibits profitability and risk, with risk and return on horizontal and vertical axes, respectively (Sharpe 1964).

The capital asset pricing model describes the connection between system risk and asset expected return. Previous research typically examined the relationship between risk and expected return using the capital asset pricing model and constructed a portfolio of risk assets (Lintner 1965). The capital asset pricing model can assist investors in risk management and portfolio optimization.

3.4 Model building

Based on the efficient market hypothesis, historical data can fully reflect current prices (Malkiel 2003). Because future information can be extracted from current information, reinforcement learning and Markov decision-making processes in financial markets may have the same perspective. In this research design, technical indicators, returns, and covariance matrices are included as internal financial information (Wang et al. 2022). As a result, the built model is fully observable and can be considered a Markov decision process (MDP) because the agent can directly observe the environment state and the agent state is equal to the environment and information states. In this study, we use three trading strategies, which are described below.

3.4.1 Actor-critic trading strategy: Advantage Actor-Critic (A2C)

To anticipate action probabilities, a policy network based on the actor-critic architecture is used. During training, probabilities are assigned to the classification distribution and samples are chosen from them. In this method, the critic network is utilized to estimate the Q function. Table 2 shows the pseudocode for the algorithm.

3.4.2 Actor-critic trading strategy: proximal policy optimization (PPO)

Proximal policy optimization (PPO), an on-policy algorithm that can be executed in discrete or continuous action space, primarily employs the combined value error term to obtain information function and policy substitution. As a result, the neural network architecture can share policy and value functions. PPO is trained with a random strategy and explores by sampling actions based on the most recent version of its random strategy. The randomness of action selection is determined by the initial conditions as well as the training process. The algorithm's pseudocode is shown in Table 3. The policy network is used to perform the observation, and the mean and standard deviation vectors are obtained. In the training phase, these mean values and standard deviations are used to create samples from a

multi-dimensional Gaussian distribution. During the testing phase, the network's average value is obtained only.

3.4.3 Actor-critic trading strategy: soft actor-critic (SAC)

Soft actor-critic (SAC) is a maximum entropy actor-critic framework. It's a non-policy approach that can be used in continuous action space. The actor aims to maximize expected reward while decreasing entropy, and in this case, acts as randomly as possible while still achieving the goal. Table 4 shows the SAC pseudo code, in which the Q function is used in the same way as the neural network, and stochastic gradient descent is used alternately to optimize the two networks. The parameterized state value function $v_\psi(s_t)$ is used, $Q_\theta(s_t, a_t)$ is the soft Q function, and the strategy $\pi_\varphi(a_t|s_t)$, ψ, θ, ϕ are network parameters. This method combines the deep Q-learning method with the stable and random actor-critic framework and is updated using the off-policy method, which performs better in continuous tasks than the previous one.

3.5 Construction of the deep reinforcement learning model

The efficient market theory demonstrated that current prices can fully reflect historical information (Malkiel 2003). Because future information may be retrieved from existing information, reinforcement learning and Markov decision-making processes have the same perspective at financial markets. Meanwhile, we are working on a comprehensive observability model. Deep learning methods A2C, PPO, and SAC use actor-critic neural network architecture to train agents labeled A2C agent, PPO agent, and SAC agent, respectively. During the interval, the parameters of the continuous model are updated and trained to identify the best approach with the highest reward. Figure 2 depicts the proposed research framework for environmental, social, and governance portfolio management using deep reinforcement learning (DRLPMESG).

Table 4 Soft Actor-Critic (SAC) algorithm pseudocode

Algorithm 3 Soft Actor-Critic (SAC)

Initialize parameter vectors $\psi, \bar{\psi}, \theta, \phi$
for each iteration **do**
  for each environment step **do**
    $a_t \sim \pi_\phi(a_t|s_t)$
    $s_{t+1} \sim P(s_{t+1}|s_t, a_t)$
    $D \leftarrow D \cup \{(s_t, a_t, r(s_t, a_t), s_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_v \hat{\nabla}_\psi J_v(\psi)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1,2\}$
    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$
    $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
  end for
end for

3.6 Environment design of deep reinforcement learning

According to the proposed DRLPMESG framework, the agent places US$100,000 on the market in the initial environment. The iShares MSCI US ESG Select Index ETF is used to create these portfolios, which are made up of five, ten, and twenty constituent equities, with different industries chosen for asset allocation. Data cleaning, feature selection, and data splitting are all processes that influence the weighting of each stock in a portfolio. Every stock in the portfolio has a positive weight, and the total weight of all stocks in the portfolio equals 1. As a dynamic strategy for portfolio trading, the weight of the stock in the portfolio is rebalanced at the end of each round (every day). Table 5 shows the pseudocode for the portfolio's deep learning model architecture.

3.7 Evaluation metrics

This study will compare the performance of different in-depth learning agents and the performance of investment back tests based on the test set. Backtesting is expected to yield the following results for the model:

(1) Annual return = $((\text{EndingValue}/\text{InitialValue})^{1/n} - 1) \times 100$,

(2) Cumulative returns = $(\text{EndingValue}/\text{InitialValue}) - 100\%$,

(3) Annual volatility =

$$\sigma_M = \left(\sqrt{\sum_{t=1}^{n} (TR_t - \overline{TR})^2} \Big/ n - 1 \right) \times \sqrt{252},$$

where n is the number of observations, σ_M is the standard deviation of the rate of return, and TR_t is the daily variation.

(4) Sharpe ratio: $S_a = E[R_a - R_b]/\sigma_a$

where R_a is the return on assets, R_b is the risk-free return, and σ_a is the standard deviation of excess return on assets.

(5) Max drawdown: (Trough Value − Peak Value)/Peak Value.

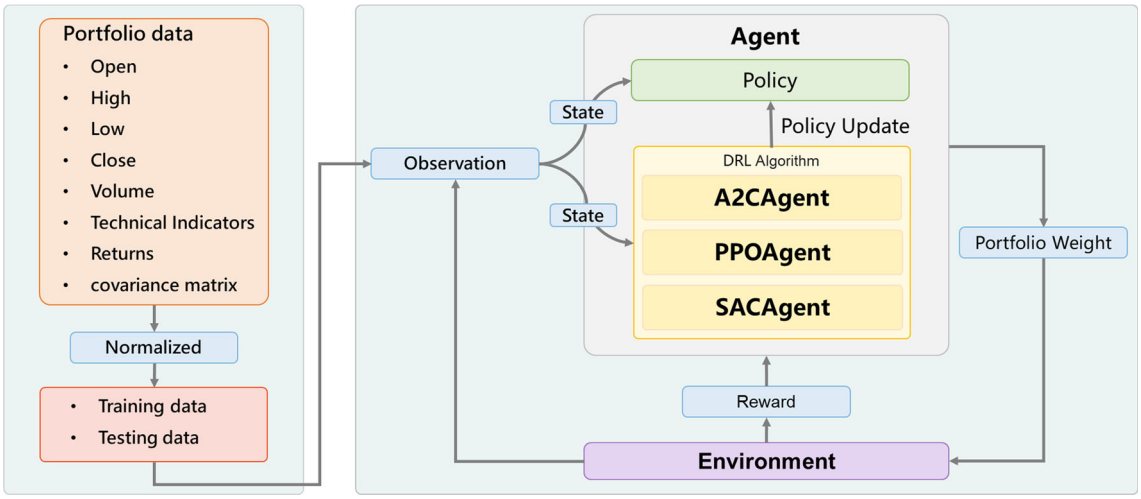


Fig. 2 Deep Reinforcement Learning for Environmental, Social, and Governance Portfolio Management (DRLPMESG)

Table 5 Pseudocode of portfolio management using deep reinforcement learning 1

Portfolio Management Using Deep Reinforcement Learning 1

1. Input: s, state space includes technical indicators and returns and covariance matrix for stocks
2. Output: Portfolio Value
3. Initialize P_0 = \$100,000, $W_0 = (\frac{1}{m}, \ldots\ldots, \frac{1}{m})$, P_0 is the initial portfolio value, W_0 *is* the initial portfolio weight, m is number of stocks in the portfolio;
4. for t = 1,…,n do

   Agent Observe a state as s and outputs a portfolio weights vector w_t as actions;

   Normalize The weight w_t to sum to 1, $\sum w_t = 1$;

   Calculate portfolio period return $R_p = \sum_{t=1}^{n} w_t \, r_t$;

   Update portfolio value $P_t = P_{t-1} \, x \, (1 + w_t^T \, r_t)$;
5. end

These results are utilized to assess the model's performance so that it can be fine-tuned to the next parameter, resulting in a more effective optimization strategy.

4 Experimental results

In this study, the efficiency frontier method is used as the static strategy. The test set is used to determine how effective back testing is. Based on a comparison of the performance of various models, portfolio management strategies, and the quantity of stocks in a portfolio, the market is used as a benchmark to determine whether implementing these strategies will result in additional excess returns.

In addition, benchmarks are the standard for measuring performance. A market index is commonly used as a benchmark in investment to evaluate portfolio performance. The benchmark will differ depending on the specific investment strategy; thus, selecting an appropriate benchmark is critical because an incorrect index will result in an incorrect benchmark (Reilly and Akhtar 1995).

4.1 Efficient frontier results

In this study, the Markowitz efficient frontier (Markowitz 1952) is used primarily to optimize the risk-reward relationship and find a configuration combination with high rewards under the same risk. Investment portfolios with the highest expected return under the same risk and the lowest risk under the same expected return are identified, as are the results for each investment portfolio. The performance of a market investment portfolio constructed using the Markowitz efficient front curve (Bodnar and Schmid 2009) with the highest expected return under the same risk is compared to that of various investment strategies.

Table 6 Backtesting results on deep reinforcement learning in portfolio performance for different agent trading strategies

Portfolio	Agent	Annual return (%)	Cumulative return (%)	Annual volatility (%)	Sharpe ratio	Stability	Max drawdown (%)
Panel A							
Portfolio 20-1	A2C agent	26.60	60.426	26.405	1.03	0.57	34.21
	PPO agent	**29.641**	**68.242**	26.968	**1.10**	0.75	32.513
	SAC agent	27.488	62.688	27.01	1.02	0.60	35.69
Portfolio 20-2	A2C agent	23.228	51.978	26.031	0.93	0.38	34.557
	PPO agent	25.588	57.867	26.015	1.01	0.58	33.922
	SAC agent	**26.675**	**58.085**	25.422	**1.03**	0.63	31.522
Portfolio 20-3	A2C agent	24.858	56.033	25.62	1.00	0.67	34.088
	PPO agent	25.689	58.12	27.078	0.98	0.70	34.314
	SAC agent	**26.995**	**61.431**	27.239	**1.01**	0.65	35.133
Panel B							
Portfolio 10-1	A2C agent	**20.234**	**44.688**	24.915	**0.87**	0.56	30.423
	PPO agent	18.259	39.945	26.298	0.77	0.23	34.954
	SAC agent	19.646	43.253	25.753	0.83	0.46	31.831
Portfolio 10-2	A2C agent	29.32	64.408	24.821	1.16	0.56	31.932
	PPO agent	**32.212**	**74.993**	25.444	**1.23**	0.58	31.791
	SAC agent	25.297	57.134	24.544	1.04	0.50	33.181
Portfolio 10-3	A2C agent	31.432	72.931	27.294	1.14	0.77	34.103
	PPO agent	27.531	62.8	27.946	1.01	0.71	34.885
	SAC agent	**34.121**	**80.094**	26.983	**1.22**	0.89	30.035
Panel C							
Portfolio 5-1	A2C agent	37.953	90.553	31.661	1.18	0.8	34.749
	PPO agent	**46.579**	**115.18**	31.7	**1.37**	0.9	32.063
	SAC agent	44.611	109.43	35.004	1.23	0.82	38.516
Portfolio 5-2	A2C agent	**22.004**	**48.966**	31.007	**0.83**	0.25	43.42
	PPO agent	21.51	47.76	28.852	0.82	0.41	37.428
	SAC agent	19.381	42.618	33.775	0.70	0.11	48.48
Portfolio 5-3	A2C agent	**38.808**	**71.289**	28.066	**1.10**	0.92	28.121
	PPO agent	26.123	59.216	27.423	0.98	0.82	28.419
	SAC agent	23.674	53.081	26.733	0.93	0.88	24.277

The significance of bold in tables used to highlight the best performing outcomes or key findings in the respective tables

4.2 Reinforcement learning test results

Table 6 shows the results of deep reinforcement learning backtesting in portfolio performance resulting from the use of different trading strategies based on A2C, PPO, and SAC Agents (i.e., three types of deep reinforcement learning methods). According to PPO Agent, portfolio 20-1 outperforms the other portfolios with 20 stocks, with an annualized return of 29.64%, a Sharpe ratio of 1.1, and a cumulative return of 68.24% as shown in Panel A of Table 6. Portfolio 5-1 outperforms the other portfolios with 5 stocks based on A2C Agent, with a 46.58% annualized rate of return, a Sharpe ratio of 1.37, and a cumulative rate of return of 115.18% as disclosed in Panel C of Table 6.

4.3 Performance of different deep reinforcement learning

Table 7 summarizes the results of the efficiency front and agents trained using three types of deep reinforcement learning methods. We reveal that self-designed active investing strategies outperformed passive investment strategies in the majority of cases. For example, SAC Agent outperformed 20-stock portfolios in two of three

Table 7 Experimental results of model and method

Portfolio	Evaluation index	Efficient frontier	A2C agent	PPO agent	SAC agent
Panel A					
Portfolio 20-1	Annual return	16.26%	26.60%	**29.64%**	27.49%
	Sharpe ratio	0.87	1.02	**1.1**	1.03
Portfolio 20-2	Annual return	15.81%	23.23%	25.59%	**26.68%**
	Sharpe ratio	0.86	0.93	1.01	**1.03**
Portfolio 20-3	Annual return	19.02%	24.86%	25.69%	**27.00%**
	Sharpe ratio	0.97	1.00	0.98	**1.01**
Panel B					
Portfolio 10-1	Annual return	15.81%	**20.23%**	18.26%	19.65%
	Sharpe ratio	0.86	**0.87**	0.77	0.83
Portfolio 10-2	Annual return	21.92%	29.32%	**32.21%**	25.30%
	Sharpe ratio	1.05	1.16	**1.23**	1.04
Portfolio 10-3	Annual return	23.77%	31.43%	27.53%	**34.12%**
	Sharpe ratio	1.00	1.14	1.01	**1.22**
Panel C					
Portfolio 5-1	Annual return	17.37%	37.95%	46.58%	**44.61%**
	Sharpe ratio	0.83	1.18	1.37	**1.23**
Portfolio 5-2	Annual return	18.02%	**22.00%**	21.51%	19.38%
	Sharpe ratio	0.69	**0.83**	0.82	0.7
Portfolio 5-3	Annual return	24.36%	**38.81%**	26.12%	23.67%
	Sharpe ratio	1.01	**1.10**	0.98	0.93

The significance of bold in tables used to highlight the best performing outcomes or key findings in the respective tables

cases as shown in Panel A of Table 7, whereas A2C Agent outperformed 5-stock portfolios in two of three cases, as shown in Panel C of Table 7. These revealed results indicate that investors can earn excess profits by employing three types of deep reinforcement learning methods, as most of their performances outperform their corresponding efficiency frontier performance in Table 7.

4.4 Benchmark trading strategy performance

This study forms these portfolios from the constituent stocks of iShares MSCI US ESG Select Index ETF constituents, so the sustainability index will be used as a benchmark to compare whether we can get excess returns in the market and outperform the market. The results are summarized in Table 8, with the gray line in the picture representing cumulative returns cumulated from the market's daily return and the green line representing the result of the backtest for the study's test set. As shown in Table 8, 7 of the 9 portfolios beat the market, with a winning rate of 78%. The experimental results show that in the majority of cases, agents can beat the market and earn excess returns, which may contradict the efficient market hypothesis (i.e., investors cannot beat the market).

4.5 Performance comparison for different numbers of portfolios

Following the evaluation and risk-return analysis, the portfolios are divided into holding 5, 10, or 20 stocks. Table 9 displays the experimental results for portfolios with varying numbers of stocks. When compared to other portfolios, the five-stock portfolio performs the best, with a 45.58% annual return and a sharp ratio of 1.37.

To demonstrate the effectiveness of our proposed method, we conduct a thorough experimental evaluation, including a comparison with the state-of-the-art methods. Table 10 presents the comparison of the proposed method with state-of-the-art methods. This analysis may cast light on the applicability of these methods to a variety of research topics, as evidenced by the citations provided below.

4.6 Discussion of computation complexity of the proposed methods

Computation complexity plays an important role in measuring of how efficiently an algorithm utilizes computational resources such as time and memory. Algorithms with lower complexity generally execute faster and consume fewer resources, leading to improved performance. In this

Table 8 Benchmark-based experimental results of the model and method

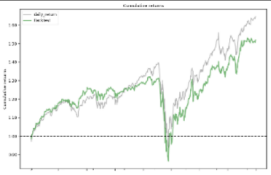
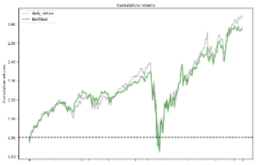
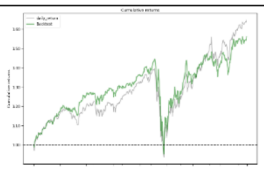
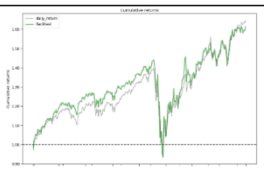
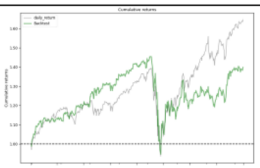
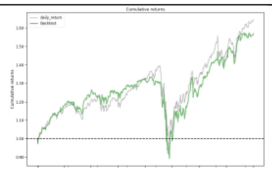
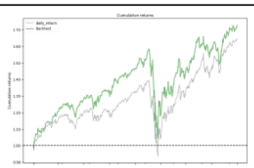
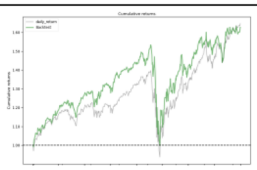
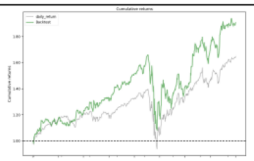
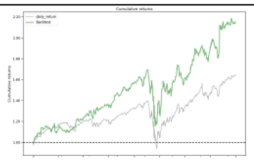
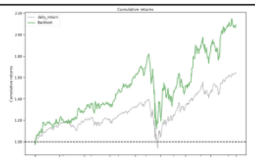
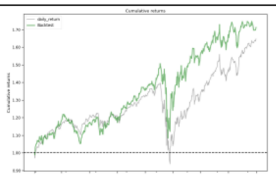
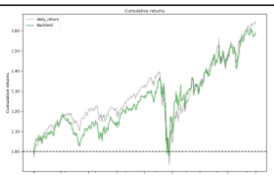
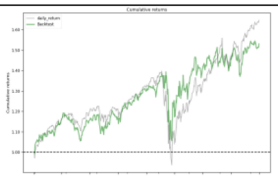
	A2CAgent	PPOAgent	SACAgent
Portfolio 20-1			
Portfolio 20-2			
Portfolio 20-3			
Portfolio 10-1			
Portfolio 10-2			
Portfolio 10-3			
Portfolio 5-1			
Portfolio 5-2			
Portfolio 5-3			

Table 9 Experimental results of different numbers of portfolio

Portfolio of 20 components		Portfolio of 10 components		Portfolio of 5 components	
Annual return	Sharpe ratio	Annual return	Sharpe ratio	Annual return	Sharpe ratio
27.49%	1.03	34.12%	1.22	**46.58%**	**1.37**

The significance of bold in tables used to highlight the best performing outcomes or key findings in the respective tables

section, we provide detailed discussion of the computation complexity of the proposed methods A2C, PPO, SAC used in this study with respect to other related methods in deep reinforcement learning as follows.

In deep reinforcement learning (RL), several methods have been proposed to address the challenge of training agents to make optimal decisions in complex environments. Three popular algorithms used in this study are Advantage Actor-Critic (A2C), Proximal Policy Optimization (PPO), and Soft Actor-Critic (SAC). We discuss the computational complexity of these methods and compare them to other related approaches in deep RL.

4.6.1 Advantage Actor-Critic (A2C)

A2C combines the advantages of both policy-based and value-based methods by using an actor-critic framework. It maintains a policy network (actor) to select actions and a value network (critic) to estimate the expected rewards. The training process in A2C is relatively straightforward, with each agent collecting experiences by interacting with the environment and updating its networks.

The computational complexity of A2C is primarily determined by the size of the network and the number of interactions with the environment. The training time scales linearly with the number of training iterations, making it suitable for parallelization. A2C can be trained efficiently on both single CPUs and distributed setups, which helps reduce the training time.

Compared to other methods, A2C is relatively lightweight in terms of computational requirements. However, its sample efficiency might be lower compared to more advanced algorithms like PPO and SAC.

4.6.2 Proximal policy optimization (PPO)

PPO is a state-of-the-art policy optimization method that addresses some limitations of earlier approaches like TRPO (Trust Region Policy Optimization). It operates by iteratively updating the policy using multiple epochs of mini-batch updates.

PPO achieves a balance between sample efficiency and computational complexity. It uses importance sampling to make efficient use of collected experiences, resulting in higher sample efficiency compared to methods like A2C. However, the computational complexity of PPO is higher due to multiple iterations over the collected data during each training epoch.

PPO's computational complexity depends on factors such as the number of training iterations, mini-batch size, and the complexity of the policy network architecture. It offers a good trade-off between efficiency and effectiveness, making it widely used in practice.

4.6.3 Soft Actor-Critic (SAC)

SAC is an off-policy RL algorithm that combines the actor-critic framework with the concept of entropy regularization. It aims to maximize the expected reward while also maximizing the policy's entropy. SAC is known for its robustness and stability during training.

The computational complexity of SAC is influenced by several factors, including the network architecture, replay buffer size, and the number of gradient updates per iteration. As an off-policy algorithm, SAC requires maintaining a replay buffer to store experiences, increasing its memory requirements compared to on-policy methods like A2C and PPO.

SAC typically requires more computational resources than A2C and PPO due to the additional entropy regularization term and the need to update both the policy and the value function. However, recent advancements in hardware and parallelization techniques have made training SAC more feasible.

In comparison to other related methods in deep RL, A2C, PPO, and SAC are generally considered to have moderate computational complexity. Advanced algorithms like the Rainbow DQN, D4PG (Distributed Distributional Deterministic Policy Gradients), or IMPALA (Importance Weighted Actor-Learner Architecture) can have higher computational requirements due to their complexity or distributed nature.

5 Conclusions

This study proposes a dynamic portfolio based on a deep reinforcement learning trading strategy to ensure stable returns even in volatile stock markets. The proposed model is a market-based model designed to outperform the market

Table 10 Comparison of the proposed method with state-of-the-art methods

References	Data set	Data period	Features setting	Deep learning method	Performance criteria	Transaction cost
Huang et al. (2020)	CSI300 index, 3 stocks in CSI500	2007–2020	OCHLV	DDPG	Average return: 0.000949 Annual Sharpe: 1.707 Annual Sortino: 2.946 MDD: 8.09%	Yes, 0.0025
Zhang et al. (2020)	50 futures contracts	2005–2019	OCHLV, MA, MACD, RSI	LSTM, DQN, PG, A2C	$E(R)$: 1.258 Std.(R): 0.976 Sharpe: 1.288, Sortino: 2.220 MDD: 0.002	Yes, 0.0001
Rundo (2019)	EUR/USD	2004–2018	OCHLV, Technical indicators	LSTM, DQN	ROI Max = 98.81% MD Max = 19.28%	No
Liu et al. (2021)	Bitcoin	2017–2020	OCHLV, technical indicators	LSTM, PPO	Profits rate = 1.46%	Yes, 0.0025
Darapaneni et al. (2020)	16 index from Indian market	2011–2019	OCHLV, SMA	Q-learning	Annualized return = 8.54% Max drawdown = 1.44%	Yes, 0.0030
Wu et al. (2020)	3 stocks from US, 3 stocks from UK, and 3 stocks from China	2008–2018	OCHLV, MA, EMA, ACD, BIAS	GRU, DQN, DPG	Return = 215.9% SR = 1.9%	No
This study	227 constituent stocks from MSCI US ESG Select Index	2000–2020	OCHLV, technical indicators	Proposed DRLPMESG framework, PPO	Annual return = 45.58% SR = 1.37	No

and even generate excess returns. As a result, not only does this study investigate various portfolio management methods, but it also recommends the best portfolio management method. Furthermore, this study investigates how many stocks should be included in a portfolio to achieve better investment performance.

5.1 Main conclusions

First, according to the benchmark method, seven out of nine portfolio groups beat the market at a winning rate of 78% in this study's experimental results. This finding confirmed that most agents can outperform the market and provide higher returns. To put it another way, the analysis and discussion of the experimental data show that actively managed investment strategies provide higher returns and outperform the market in general.

Second, we also find that self-designed active portfolio management strategies outperform passive portfolio management strategies. For example, SAC agent outperforms 20-stock portfolios with a cumulative rate of return of 115.18%, while A2C agent outperforms 5-stock portfolios with an annualized rate of return of 46.58% and a Sharpe ratio of 1.37, respectively. These findings are somewhat different from previous theories because passively managed investment strategies outperform actively managed investment strategies, and that investors would neither make excessive profits nor beat the market.

Finally, while investigating how many stocks to hold in a portfolio to attain the best performance, we discovered that a portfolio consisting of five stocks has the best performance. Notably, holding five stocks in a portfolio can result in higher returns and even lower risk in the portfolio as measured by risk and return. This finding contradicts previous research findings, which contend that multiple stocks must be held in the portfolio to reap decentralized benefits. This result, however, has not previously been described.

5.2 Study strength and contributions

The experimental results of this study may contribute to our understanding of portfolio management in a variety of ways, as well as serve as a foundation for future research.

We argue that the following are the main contributions of this study. First, by employing the deep reinforcement learning for portfolio management on environmental, social, and governance (DRLPMESG) model, this study not only proposes various active investment strategies for portfolio management, but it also provides a thorough evaluation of portfolio performance. Second, we show that the majority of the experimental results outperform the market and generate excess profits for the DRLPMESG model as an investment strategy and active portfolio management, indicating that the proposed model can effectively generate remuneration. Third, we demonstrate that 5-stock portfolios can generate high returns using portfolio dynamic trading strategies based on deep reinforcement learning, which may not only shed new light on how to improve portfolio management but also provide a theoretical foundation for future research on the issue of how many stocks should be held in a portfolio.

5.3 Research implications

This study applies deep reinforcement learning to portfolio management, analyzes the relationship between risk and expected return, and randomly selects low-correlation assets from various industries to form a portfolio. In this study, the deep reinforcement learning model is used to observe the financial market and formulate a dynamic strategy. The research results are obtained by thoroughly analyzing various methods. These findings have the following important research implications for investors and market observers.

First, the designed and formulated investment strategy focuses on firms that aim to achieve stable and long-term profitability, as well as those that contribute to social sustainability and have excellent ESG evaluation performance. To design in-depth reinforcement learning, a sustainable investment strategy with active portfolio management is proposed. As a result, ESG firms are the portfolio's targets, and the results show that investing in these firms would generate excess returns, implying that incorporating ESG stocks into portfolios may have a positive impact on portfolio performance.

Second, this study investigates how many stocks should be included in a portfolio to generate a higher return. According to the study, a portfolio of five stocks (i.e., a small number of stocks) not only simplifies management but also provides investors with excess returns, which may contradict the premise that a portfolio should hold that many stocks. We then conclude that this unexpected finding could be due to the fact that managing multiple stocks in a portfolio can be difficult and may increase management costs for investors using self-designed strategies.

Thirds, this study's robustness test uses the 2-year period from 2019 to 2020 to ensure long-term and stable profits under tolerable risks and a fixed portfolio held over time. As such, updating agent weights every day (round) after observing the environment and providing different model options is a viable trading strategy, implying that dynamic strategies, as opposed to previous static strategies, can respond to extreme market changes faster and are more adaptable in application.

5.4 Future research and limitation

In this study, portfolios are divided into 5, 10, and 20 stocks. The findings reveal that holding five stocks in a portfolio can result in better portfolio performance. However, we suspect that the accuracy will decrease as fewer stocks are only added to a portfolio. Thus, whether portfolios comprised of fewer than five stocks would produce accurate results with significantly better performance based on the proposed framework would be worthwhile for future research.

Although other components of the financial market, such as news, public opinion, current affairs, and other information can be included, the scope of this study only covers quantities, rewards, risks, benchmarks, and models, which may be the limitation of this study. As a result, future research can use the offered model to include more of the aforementioned components and investigate the extent of getting closer to financial internal information.

Additionally, we argue that there is still plenty of room for further research as follows. To begin, this study may look into the relationship between attribute noise and classification accuracy, the influence of noise at different attributes (Zhu and Wu 2004), and feasible ways for dealing with attribute noise in financial markets utilizing deep reinforcement learning. Furthermore, this research can help to broaden our understanding of attribute noise and its impact on classification accuracy in the context of portfolio optimization in financial markets. Moreover, this study may delve deeper into the complex relationship between attribute noise and the prediction power of deep reinforcement learning algorithms used in portfolio management methods. Besides, the performance of several noise-handling approaches such as noise correction, instance elimination, and feature selection could be further evaluated in this study. In other words, we argue that the aforementioned concerns may broaden the scope for future research, provide a better knowledge for enhancing the robustness of AI-driven portfolio management systems, and contribute more to financial technology and the broader field of artificial intelligence.

Funding This research was supported by the Ministry of Science and Technology (MOST), Taiwan [110-2410-H-305-013-MY2] and National Taipei University (NTPU), Taiwan [111-NTPU_ORDA-F-001 and 111-NTPU_ORDA-F-003].

Availability of data and materials The datasets used and/or analyzed during the current study are available from the first author on reasonable request at myday@gm.ntpu.edu.tw.

Declarations

Conflict of interests The authors declare that they have no competing interests.

References

Aggarwal S, Aggarwal S (2017) Deep investment in financial markets using deep learning models. Int J Comput Appl 162:40–43

Ahmed S, Alshater MM, ElAmmari A, Hammami H (2022) Artificial intelligence and machine learning in finance: a bibliometric review. Res Int Bus Finance 61:101646

Al-Aradi A, Jaimungal S (2021) Active and passive portfolio management with latent factors. Quant Finance 21:1437–1459

Alexander C (2008) Market risk analysis, practical financial econometrics. Wiley

Arulkumaran K, Deisenroth MP, Brundage M, Bharath AA (2017) Deep reinforcement learning: a brief survey. IEEE Signal Process Mag 34:26–38

Betancourt C, Chen W-H (2021) Deep reinforcement learning for portfolio management of markets with a dynamic number of assets. Expert Syst Appl 164:114002

Bodnar T, Schmid W (2009) Econometrical analysis of the sample efficient frontier. Eur J Finance 15:317–335

Castiglioni I, Rundo L, Codari M et al (2021) AI applications to medical images: From machine learning to deep learning. Phys Med 83:9–24

Chandra P (2017) Investment analysis and portfolio management. McGraw-Hill Education, New York

Charpentier A, Elie R, Remlinger C (2023) Reinforcement learning in economics and Finance. Comput Econ 62:425–462

Chen YF, Huang SH (2021) Sentiment-influenced trading system based on multimodal deep reinforcement learning. Appl Soft Comput 112:107788

Chen R, Ren J (2021) Do AI-powered mutual funds perform better? Finance Res Lett 47:102616

Chen J, Luo C, Pan L, Jia Y (2021) Trading strategy of structured mutual fund based on deep learning network. Expert Syst Appl 183:115390

Choueifaty Y, Coignard Y (2008) Toward Maximum Diversification. J Portf Manag 35:40–51

Craja P, Kim A, Lessmann S (2020) Deep learning for detecting financial statement fraud. Decis Support Syst 139:113421

Da Silva AS, Lee W, Pornrojnangkool B (2009) The Black-Litterman model for active portfolio management. J Portf Manag 35:61

Darapaneni N, Basu A, Savla S, et al (2020) Automated portfolio rebalancing using Q-learning. In: 2020 11th IEEE annual ubiquitous computing, electronics & mobile communication conference (UEMCON). IEEE, pp 0596–0602

DeMiguel V, Gil-Bazo J, Nogales FJ, Santos AA (2021) Can machine learning help to select portfolios of mutual funds?, SSRN

Deng Y, Bao F, Kong YY et al (2017) Deep direct reinforcement learning for financial signal representation and trading. IEEE Trans Neural Netw Learn Syst 28:653–664

Escrig-Olmedo E, Muñoz-Torres MJ, Fernandez-Izquierdo MA (2010) Socially responsible investing: sustainability indices, ESG rating and information provider agencies. Int J Sustain Dev 2:442–461

Evans JL, Archer SH (1968) Diversification and the reduction of dispersion: an empirical analysis. J Finance 23:761–767

Huang G, Zhou X, Song Q (2020) Deep reinforcement learning for portfolio management. arXiv preprint arXiv:201213773

Ieda M, Fujino N, Sasaki H (2019) Active portfolio management with conditioning information. J Investig 28:51–65

Kocmanová A, Dočekalová M (2012) Construction of the economic indicators of performance in relation to environmental, social and corporate governance (ESG) factors. Acta Universitatis Agriculturae Et Silviculturae Mendelianae Brunensis 60:195–206

Liang Z, Chen H, Zhu J, et al (2018) Adversarial deep reinforcement learning in portfolio management. arXiv preprint arXiv:180809940

Lim QYE, Cao Q, Quek C (2022) Dynamic portfolio rebalancing through reinforcement learning. Neural Comput Appl 34:7125–7139

Lintner J (1965) The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. Rev Econ Stat 47:13–37

Liu FR, Li Y, Li BT et al (2021a) Bitcoin transaction strategy construction based on deep reinforcement learning. Appl Soft Comput 113:107952

Lucarelli G, Borrotti M (2020) A deep Q-learning portfolio management framework for the cryptocurrency market. Neural Comput Appl 32:17229–17244

Ma YL, Han RZ, Wang WZ (2021b) Portfolio optimization with return prediction using deep learning and machine learning. Expert Syst Appl 165:113973

Malkiel BG (2003) The efficient market hypothesis and its critics. J Econ Perspect 17:59–82

Markowitz H (1952) Portfolio selection. J Finance 7:77–91

Meng TL, Khushi M (2019) Reinforcement learning in financial markets. Data 4:110

Ozbayoglu AM, Gudelek MU, Sezer OB (2020) Deep learning for financial applications: a survey. Appl Soft Comput 93:106384

Peterson S (2012) Active portfolio management. In: Investment theory and risk management, pp 187–196

Pham U, Luu Q, Tran H (2021) Multi-agent reinforcement learning approach for hedging portfolio problem. Soft Comput 25:7877–7885

Reilly FK, Akhtar RA (1995) The benchmark error problem with global capital markets. J Portf Manag 22:33

Rundo F (2019) Deep LSTM with reinforcement learning layer for financial trend prediction in FX high frequency trading systems. Appl Sci 9:4460

Sharpe WF (1964) Capital asset prices: a theory of market equilibrium under conditions of risk. J Finance 19:425–442

Shi S, Li JJ, Li GH et al (2022) GPM: a graph convolutional network based reinforcement learning framework for portfolio management. Neurocomputing 498:14–27

Stoilov T, Stoilova K, Vladimirov M (2021) Application of modified Black-Litterman model for active portfolio management. Expert Syst Appl 186:115719

Sutton RS, Barto AG (1998) Reinforcement learning: an introduction. MIT Press, Cambridge

Taghian M, Asadi A, Safabakhsh R (2022) Learning financial asset-specific trading rules via deep reinforcement learning. Expert Syst Appl 195:116523

Théate T, Ernst D (2021) An application of deep reinforcement learning to algorithmic trading. Expert Syst Appl 173:114632

Torrente ML, Uberti P (2022) A rescaling technique to improve numerical stability of portfolio optimization problems. Soft Comput. https://doi.org/10.1007/s00500-021-06543-1

Vo NNY, He XZ, Liu SW, Xu GD (2019) Deep learning for decision making and the optimization of socially responsible investments and portfolio. Decis Support Syst 124:113097

Wang HN, Liu N, Zhang YY et al (2020) Deep reinforcement learning: a survey. Front Inf Technol Electron Eng 21:1726–1744

Wang JZ, Zhang HP, Luo H (2022) Research on the construction of stock portfolios based on multiobjective water cycle algorithm and KMV algorithm. Appl Soft Comput 115:108186

Weng LG, Sun XD, Xia M et al (2020) Portfolio trading system of digital currencies: a deep reinforcement learning with multidimensional attention gating mechanism. Neurocomputing 402:171–182

Wu X, Chen H, Wang J et al (2020) Adaptive stock trading strategies with deep reinforcement learning methods. Inf Sci 538:142–158

Wu ME, Syu JH, Lin JCW, Ho JM (2021) Portfolio management system in equity market neutral using reinforcement learning. Appl Intell 51:8119–8131

Xu ZY, Zhang J, Wang JY, Xu ZM (2020) Prediction research of financial time series based on deep learning. Soft Comput 24:8295–8312

Yu XM, Wu WJ, Liao XC, Han Y (2023) Dynamic stock-decision ensemble strategy based on deep reinforcement learning. Appl Intell 53:2452–2470

Yun H, Lee M, Kang YS, Seok J (2020) Portfolio management via two-stage deep learning with a joint cost. Expert Syst Appl 143:113041

Zhang Z, Zohren S, Roberts S (2020) Deep reinforcement learning for trading. J Financ Data Sci 2:25–40

Zhu X, Wu X (2004) Class noise vs. attribute noise: a quantitative study. Artif Intell Rev 22:177

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.